

ALEXANDRE S. AVARO

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EDUCATION

Ph.D. in Mechanical Engineering, Stanford University, GPA 4.1/4.0 Sep 2020 - Aug 2025
Thesis title: CRISPR assays and microfluidic devices for quantitative diagnostic methods.
Supervisors: Juan G. Santiago & Andrew D. Griffiths.

Master of Science in Mechanical Engineering, Stanford University, GPA 4.07/4.0 Sep 2020 - Jun 2022
Depth: Fluid Mechanics. Breadth: Solid Mechanics.
Relevant Coursework: Fluid Mechanics, Fluid Flow in Microdevices, Turbulence, Compressible Flows, Multiphase Flows, Continuum Mechanics, Numerical & Asymptotic Methods, Finite Elements Analysis, Signal Processing.

Diplôme d'ingénieur, CentraleSupélec (Université Paris-Saclay), GPA 4.0/4.0 Sep 2018 - Jun 2022
Major: Aeronautics, Space, Transport. Bachelor in Engineering completed in Sep 2019.
Relevant Coursework: Fluid Mechanics, Solid Mechanics, Multiphysics Coupling, Material Sciences, Electromagnetism, Partial Differential Equations, Convergence-Integration-Probabilities, Optimization, Statistics, Algorithmics.

Classes Préparatoires aux Grandes Écoles PCSI-PC*, Lycée Louis le Grand, GPA 4.0/4.0 2016 - 2018

EXPERIENCE

Postdoctoral Researcher, École Polytechnique & Institut Pasteur, CNRS Sep 2025 - Present
LadHyX & Physical Microfluidics and Bioengineering team, under the supervision of Profs. Charles Baroud and Gabriel Amselem. *Palaiseau, France*

- Hydrogel-based actuators for mechanobiology.
- Supervised two graduate interns and one undergraduate intern.

Research Engineer, ESPCI, CNRS Jan 2023 - Aug 2025
Biochemistry Laboratory, under the supervision of Prof. Andrew D. Griffiths. *Paris, France*

- Control and dynamics of DNA in multifurcation microfluidic networks using isotachophoresis.
- CRISPR-based diagnostics for trisomy 21. Developed high-throughput sequence screening pipeline for CRISPR Cas12 using droplet-based microfluidics and machine learning.
- Co-supervised one undergraduate intern (C. Dupire).

Graduate Research Assistant, Stanford University Oct 2020 - Aug 2025
Stanford Microfluidics Laboratory, under the supervision of Prof. Juan G. Santiago. *Stanford, CA*

- Developed and experimentally validated isotachophoresis/electrophoresis simulators [CAFES](#) and [BEAN](#).
- Developed novel techniques for nucleic acid diagnostics combining CRISPR and microfluidics.
- Studied fundamental theory of CRISPR kinetics and its link to the limit of detection of CRISPR-based assays for molecular diagnostics.
- Supervised three undergraduate interns (L. Agrawal, L. Brown, A. Schwarzbach).

Teaching Assistant, Stanford University Jan 2022 - Jun 2025
ME360: Physics of Microfluidics, taught by Prof. Juan G. Santiago. (graduate level) *Stanford, CA*
ME354: Experimental Methods in Fluid Mechanics, taught by Prof. Juan G. Santiago. (graduate level)
ME351B: Fluid Mechanics, taught by Prof. Juan G. Santiago. (graduate level)
CME102: Ordinary Differential Equations, taught by Prof. Hung Le. (undergraduate level)

Teaching Assistant, CentraleSupélec Feb 2019 - May 2019
Quantum and Statistical Physics, taught by Prof. Jean-Michel Gillet. (undergraduate level) *Gif-sur-Yvette, France*

PEER-REVIEWED PUBLICATIONS

Leon V. Gebhard, **Alexandre S. Avaro**, Gabriel Amselem, and Charles N. Baroud, “Designing single-layer PDMS devices for micron to millimeter-scale deformations”, under review at *Lab on a Chip*, **2026**.

Qi Jiang, **Alexandre S. Avaro**, Hyunjun Bae, Angelina Sorensen, and Juan G. Santiago, “Experimental Methods for CRISPR Enzyme Assays with Fluorescence Readout”, under review at *ACS Sensors*, **2026**.

Qi Jiang, Ashwin Ramachandran, **Alexandre S. Avaro**, Diego A. Huyke, and Juan G. Santiago, “Reaction kinetics of CRISPR *trans*-cleavage controlled using isotachophoresis”, *Analytical Chemistry*, 97, 50, 27646-27653, **2025**. [Link](#).

Alexandre S. Avaro, Shahab Mirjalili, Andrew D. Griffiths, and Juan G. Santiago, “Microfluidic networks using isotachophoresis”, *Proceedings of the National Academy of Sciences*, 122, 45, e2511724122, **2025**. [Link](#).

Alexandre S. Avaro and Juan G. Santiago, “Engineering guidelines for CRISPR diagnostics”, *Chemical Communications*, 61, 95, 18756-18772, **2025**. [Link](#). Featured on the inside front [cover](#) of *Chemical Communications*.

Alexandre S. Avaro, Andrew D. Griffiths, and Juan G. Santiago, “Degradation of Reporter Molecules Imposes a Fundamental Limit of Detection on CRISPR Diagnostics”, *Analytical Chemistry*, 97, 25, 13192-13200, **2025**. [Link](#).

Qi Jiang, Xuehao Zang, Yilu Wang, **Alexandre S. Avaro**, Diego A. Huyke, and Juan G. Santiago, “A Three-Dimensional Electrokinetic Microfluidic Device Embedded within a Thermal Cycler Tube”, *Lab on a Chip*, 25, 3962-3970, **2025**. [Link](#).

Alexandre S. Avaro,[†] Adar Schwarzbach,[†] Amit Jangra, Supreet S. Bahga, and Juan G. Santiago, “Highly Parallel Simulation Tool for the Design of Isotachophoresis Experiments”, *Analytica Chimica Acta*, 1337, 343553, **2025**. [Link](#). Featured on the outside front [cover](#) of *Analytica Chimica Acta*. [†]: Contributed equally.

Diego A. Huyke,[†] **Alexandre S. Avaro**,[†] Thomas Kroll, and Juan G. Santiago, “Stream Lamination and Rapid Mixing in a Microfluidic Jet for X-ray Spectroscopy Studies”, *Flow*, 3, E25, **2023**. [Link](#).

Alexandre S. Avaro and Juan G. Santiago, “A Critical Review of Microfluidics-Driven CRISPR Diagnostics”, *Lab on a Chip*, 23, 938-963, **2023**. [Link](#).

Alexandre S. Avaro and Juan G. Santiago, “Uncertainty Quantification of Michaelis–Menten Kinetic Rates and Its Application to the Analysis of CRISPR-Based Diagnostics”, *Angewandte Chemie International Edition*, 61, 45, e202209527, **2022**. [Link](#).

Charles Blanluet, Diego A. Huyke, Ashwin Ramachandran, **Alexandre S. Avaro**, and Juan G. Santiago, “Detection and Discrimination of Single Nucleotide Polymorphisms by Quantification of CRISPR-Cas Catalytic Efficiency”, *Analytical Chemistry*, 94, 43, 15117–15123, **2022**. [Link](#).

Alexandre S. Avaro, Yixiao Sun, Kaiying Jiang, Supreet S. Bahga, and Juan G. Santiago, “Web-Based Open-Source Tool for Isotachophoresis”, *Analytical Chemistry*, 93, 47, 15768–15774, **2021**. [Link](#).

CONFERENCES

TALKS

Alexandre S. Avaro, Leon V. Gebhard, Gabriel Amselem, and Charles Baroud, “Locally recirculating flow at near-zero Reynolds number”, *Microfluidic Horizons 2026*, Padua, Italy, **2026**. [Link](#).

POSTER PRESENTATIONS

Alexandre S. Avaro, Andrew D. Griffiths, and Juan G. Santiago, “Electrokinetic Device for Parallel Enzymatic DNA Amplification Assays Compatible with Raw Input Sample”, *The 28th International Conference on Miniaturized Systems for Chemistry and Life Sciences - Micro-Total Analysis Systems (μ TAS)*, Montréal, Canada, **2024**. [Link](#). Awarded CBMS travel grant.

Alexandre S. Avaro, Andrew D. Griffiths, and Juan G. Santiago, “Electric-field driven microfluidic devices for affinity-based protein separation”, *Single-Molecule Sensors and NanoSystems International Conference*, Barcelona, Spain, **2023**. [Link](#).

Alexandre S. Avaro and Juan G. Santiago, “Uncertainty Quantification of Michaelis–Menten Kinetic Rates and Its Application to the Analysis of CRISPR-Based Diagnostics”, *Microfluidics Global Exposition*, Paris, France, **2022**.

PATENTS

Alexandre S. Avaro and Juan G. Santiago. “Enhanced Calibration Method for CRISPR-Based Diagnostics”. Provisional Patent Application. [Link](#).

Alexandre S. Avaro and Juan G. Santiago. “Apparatuses and Methods involving Spatiotemporal Sequencing of Droplet Assays”. International Patent Application [WO2024148025A1](#) (patent pending).

Juan G. Santiago, **Alexandre S. Avaro**, Ashwin Ramachandran, Diego A. Huyke, and Charles Blanluet. “Enzyme Kinetics Analyses of CRISPR Endonucleases”. International Patent Application [WO2023201275A1](#) (patent pending, licensed).

SKILLS

Experimental skills: Microfluidic system design and control, microfabrication and prototyping (PDMS chip fabrication, soft lithography, 3D printing using photolithography, CNC machining), epifluorescence microscopy, quantitative image processing, electrokinetic system control, biomolecular techniques (sample preparation, molecular biology experimental design: PCR, qPCR, LAMP, IVT, CRISPR-Cas12), data analysis and processing.

Software: Python, MATLAB, Inkscape, Microsoft Office Suite, and $\text{\LaTeX}$. Basic proficiency with JavaScript (React), keras, Mathematica, and Adobe InDesign.

CAD and Simulation software: Proficient with COMSOL Multiphysics, SOLIDWORKS, AutoCAD, Fusion360, and Rhino/Grasshopper.

Languages: French (native), English (fluent, TOEFL iBT 112/120, IELTS 8.0), German (B2+ CEFR).

OTHER PROJECTS

Peer reviewer for archived journals. *Analytica Chimica Acta* (2025), *Lab on a Chip* (2025).

Correlation between cone snail venom toxicity and shell resistance. Supervised by Profs. Véronique Aubin (CentraleSupélec) and Nicolas Puillandre (Museum of Natural History of Paris). Performed and analysed three-point mechanical bending tests of more than 100 snail shells.

Mechanical test devices database for CNRS. Supervised by Dr. Eva Héripré (CNRS). Built a database of mechanical test devices of the CNRS (French National Centre for Scientific Research). Developed a platform to register all test devices.

Investigation of triboelectric polymer design. Supervised by Prof. Pierre Jehel (CentraleSupélec). Manufactured and studied the mechanical response of several polymers with triboelectric properties.

OTHER ACTIVITIES

- Served as judge at iGEM, a worldwide competition in molecular biology with 400+ undergraduate and post-graduate teams in 2023 and 2024.
- Participated in science popularization events in ESPCI (Nuit Européenne des Chercheur·es).
- Served as president of the weekly newspaper of CentraleSupélec, printed at more than 250 copies per week. Staged and acted in two theater comedies. Served as Secretary General of the CentraleSupélec debating society.
- Student representative in CentraleSupélec. Participated in discussions on the new curriculum of CentraleSupélec.
- Volunteered as a tutor for undergraduate students in Mathematics and Physics (Lycée Lakanal).
- Won the Stanford Kennedy Residences 2020 “Iron Chef” cooking competition.